

College of Computing and Information Technology
Department of Information Technology

Course Specification - Database Systems

Document Code: TP-153 / CPIT320 / 2026

A. Course Identification

Field	Details
Course Code	CPIT320
Course Title	Database Systems
Reference Term	Spring 2026
Primary Assessment	Midterm Examination

B. Course Learning Outcomes (CLOs)

ID	Course Learning Outcome
CLO1	Explain relational database concepts, keys, and integrity constraints.
CLO2	Apply normalization principles to improve relational schema design.
CLO3	Construct SQL queries for data retrieval and aggregation.
CLO4	Analyze transaction management, concurrency-control problems, and recovery mechanisms.

C. Course Topics

ID	Topic	Hours
T1	Relational Model, Keys, and Integrity Constraints	6
T2	Functional Dependencies and Normalization	8
T3	SQL Queries and Aggregation	8
T4	Transactions and ACID Properties	5
T5	Concurrency Control and Recovery	5

D. Assessment-to-CLO Alignment

Assessment Activity	Target CLOs
Midterm Exam	CLO1, CLO2, CLO3, CLO4

E. Midterm Examination Topic Scope

Assessment Activity	Required Topic Scope
Midterm Exam	T1, T2, T3, T4, T5

Course Coordinator

Department Chair